

TED (10)-5011

**MODEL**

REG.NO.....

REVISION – 2010

SIGNATURE.....

SIXTH SEMESTER DIPLOMA MODEL EXAMINATION IN CIVIL ENGINEERING

**GROUND IMPROVEMENT TECHNIQUES**

(Maximum marks : 100)

[time : 3 hrs]

**PART – A**

(Answer all questions )

(Each question carries 2 marks)

- I. Answer the following questions in one or two sentences
1. Define permeability of soil.
  2. Give classification of soil based on particle size in mechanical stabilisation.
  3. Define geosynthetics.
  4. Define consolidation of soil.
  5. List down the shear stress parameters of soils. (5x2 = 10)

**PART – B**

(NAXIMUM MARKS – 5x6 = 30)

(ANSWER ANY FIVE QUESTIONS)

- II.
1. List down some of the in-situ methods of ground improvements.
  2. Draw the schematic arrangement of a open sump method for dewatering.
  3. Define properties of grout.
  4. Explain the procedure for mechanical stabilisation.
  5. Explain the stability conditions applied for design of reinforced earth wall.
  6. Compare compaction and consolidation of soil.
  7. Explain the procedure of conducting consolidation test.
- (5x6= 30)

**PART – C**

(ANSWER ONE FULL QUESTION FROM EACH UNIT)

(MAXIMUM MARKS – 4x15= 60)

**Unit. I**

- III. (a) What is the principle of dewatering by well points and sumps? (7)

(b) Give the precautions taken for electro- osmosis method of dewatering. (8)

OR

IV. (a) Explain the suitability of dewatering by deep well method. (6)

(b) Recommend and justify different methods for dewatering for following site conditions.

1) Water table to be lowered by 17 m in sandy soil.

2) Shallow cut for foundation with water table above ground level.

3) Silty clay and fine sand for considerable depth.

4) Soil at site is silty clay with peat. (9)

#### UNIT – II

V. (a) Give the types of soil-Bitumen with their suitability for various site conditions. (9)

(b) Explain jet grouting and its applications in civil engineering. (6)

OR

VI. (a) Explain the method of calcium chloride stabilisation. (6)

(b) What are the types of rollers used in compacting soil surfaces. (8)

#### UNIT III

VII. (a) Write down the procedure for reinforced earth wall construction (9)

(b) Explain the possible failure modes of reinforced soil formations. (6)

OR

VIII. (a) Give any four applications of geosynthetics in civil engineering. (8)

(b) Explain the design considerations of reinforcing soil. (7)

#### UNIT IV

IX. (a) Explain the importance of e- log P curve for consolidation test. (7)

(b) Explain the practical application of forced consolidation for soils. (8)

OR

X. (a) Mention various stages of consolidation of soils. (6)

(b) Explain the procedure for direct shear test. (8)

